# Silent-Failure Study

Status as of 22 August 2026

## Where you are: **the baseline is locked.** Week 1 begins.

The measurement of your own engine is finished and reproducible. Everything remaining is about the other two engines.

## Done

- ✓ **Engine frozen** at `a8dc2b2`, tag `v2.1.3` (`FREEZE.md`).
- ✓ **Baseline measured and re-verified.** 55 cases: 44 pass, **0 silent**, 1 error, 10 timeout; 45 adjudicated.
- ✓ **Reproduction confirmed** — bit-identical to the 10 August run. Zero status changes, no metric changed at all. This also forecloses "could that be harness noise?" against the week-2 comparison.
- ✓ **Third engine chosen:** Drake. ✓ **Documented:** freeze record (16 pp) + fixes writeup (5 pp).

## Next — Week 1: adapters

*This is the hard part and where all the risk lives. Budget the whole week.*

- ☐ **1. Library-independent $N$-link reference** (`numpy` only). Closed-form $M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = 0$. **Do this first** — it is the ground truth the SymPy column needs, since a SymPy oracle checking a SymPy engine is near self-consistency. Doubles as the gate case.
- ☐ **2. SymPy adapter.** Pendulum chain via `sympy.physics.mechanics`. Already installed, so it is the smaller of the two.
- ☐ **3. Install Drake in Ubuntu WSL.** Not native Windows. The heavier install — start it early, in parallel with steps 1–2.
- ☐ **4. Drake adapter.** Same chain, same knobs.
- ☐ **5. The gate:** all three engines agree with the reference on a hand-checked $N = 2$ case. Do not proceed past this until they do.

**Two decisions owed in week 1:** whether the `loops` axis enters the cross-engine claim (three constrained cases, currently zero strong-oracle coverage); and whether to pin one integrator family and tolerance across all three engines, since otherwise cross-engine energy drift measures the integrator rather than the engine.

**Then — Week 2:** sweep all three, build the comparison table, write.

## Rules in force until the study closes

- ! **The engine does not change.** Fixing MechanicsDSL mid-study is what makes the numbers meaningless. If it moves off `a8dc2b2`, the baseline is void and the sweep must be repeated.
- ! **The scoring path and case matrix are frozen** — but **the harness may grow**. Adapters, dispatch, and new oracles are additive. The gate: re-run the frozen matrix at `a8dc2b2` and confirm it still returns 44/0/1/10 bit-identically. Extending coverage is allowed; revising judgement is not.
- ! **Quote numbers only from `RESULTS.md`.** The 28 July report describes a pre-fix engine and must never be cited as current.

## Known limit on the headline

| Pathway | Cases | Timed out | Adjudicated | Strong oracle |
|---|---|---|---|---|
| lagrangian | 26 | 3 | 23 | 22 |
| hamiltonian | 19 | 5 | 14 | 0 |
| constrained | 10 | 2 | 8 | 0 |
| **Total** | **55** | **10** | **45** | **22** |

**33 of 55 cases (60%) carry no independent derivation check**, and 7 of those returned no answer at all. The oracle gap and the timeout wall fall in the *same* place. State the result **by pathway**, never pooled.

## At a glance

| | |
|---|---|
| Pin | `a8dc2b2` tag `v2.1.3` (`git checkout v2.1.3`) |
| Headline | Zero silent failures in 45 adjudicated cases |
| Caveat | Independent oracle covers 22 of those 45 |
| Timeout | 180 s per case |
| Engines | MechanicsDSL ✓ SymPy ☐ Drake ☐ |
| Paper | Short or workshop paper. *Not* a registered report. |
| Plan | `stress_suite/SCOPE.md` |
| Pin record | `stress_suite/FREEZE.md` |
| Numbers | `stress_suite/RESULTS.md` |
| Why they changed | `stress_suite/FIXES.md` |
| Repository | `Desktop\mechanicsdsl-main` |

Disclose in the paper: you maintain one of the engines measured, and AI assistance was used. Both are permitted; concealing either is not.